ModelIt!

Student Activity Pack

How Soap Actually Destroys Viruses

The Chemical Reaction That Saves Millions of Lives Every Year

Grade 5 | NGSS 5-PS1-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you wash your hands with soap? What does soap actually do?

2 Which is MORE effective: water only, soap only, or soap + water together?

3 What do you think happens to viruses when soap touches them?

4 Why do you think we're told to wash for 20 seconds?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Soap Concentration | Water Present | Soap Effectiveness | Virus Survival Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Water Only

Soap at 0%, Water at 100%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Dry Soap

Soap at 100%, Water at 0%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Proper Handwashing

Both at HIGH

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Scrubbing Time — More time = more soap contact with surfaces

Water Temperature — Warm water helps soap molecules move

Surface Area Covered — Missed spots still harbor viruses

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

How does soap destroy viruses at the molecular level? Use evidence from your simulation to support your answers.

1. **Why isn't rinsing with water alone enough to kill viruses?**

2. **Why is hand sanitizer (alcohol) useful but different from soap?**

3. **What percentage of viruses survive proper handwashing?**

4. **If soap is so effective, why do diseases still spread?**

5. **How did understanding soap chemistry help during COVID-19?**

8 STEM Challenge: Design a Handwashing Campaign

ModelIt!

YOUR ENGINEERING MISSION

Design a school campaign to improve handwashing and reduce illness.

THE SCENARIO

Your school has had several flu outbreaks. Design a science-based campaign to teach effective handwashing.

Engineering Design Thinking

1. WHAT VISUALS WILL SHOW WHY SOAP WORKS?

2. HOW WILL YOU DEMONSTRATE PROPER TECHNIQUE?

3. WHERE WILL YOU DISPLAY YOUR CAMPAIGN?

4. HOW WILL YOU MEASURE IF IT'S WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Public Health Scientists and Virologists study how diseases spread and how to stop them. They earn $70,000-$115,000/year.